# Remostreig

## Tema 2: Resampling for Simulation and Montecarlo method

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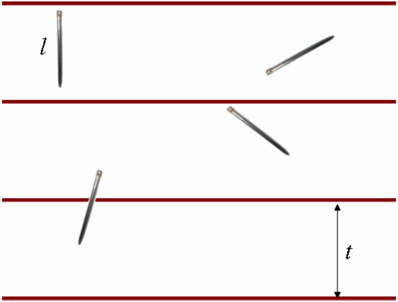
# Experiment

**Sometimes we want to do an experiment...**

# Experiment with needles

What is the probability that the needle will cross some line?




Answer: $P_x = \dfrac{2l}{\pi t} \Rightarrow \pi = \dfrac{2l}{P_x t}$, then we can approximate $\pi$ empirically with an experiment.

# Experiment

**Sometimes we want to do computations stochastic processes...**

# Experiment with $e$ number

IF YOU PICK A UNIFORMLY RANDOM REAL NUMBER IN $[0,1]$ AND REPEAT THIS UNTIL THE SUM OF THE NUMBERS PICKED IS GREATER THAN $1$, YOU'LL ON AVERAGE PICK $e \simeq 2.718\ldots$ NUMBERS!

# Experiment

**And we can solve them analytically...**

# Experiment with $e$ number

> IF YOU PICK A UNIFORMLY RANDOM REAL NUMBER IN $[0,1]$ AND REPEAT THIS UNTIL THE SUM OF THE NUMBERS PICKED IS GREATER THAN $1$, YOU'LL ON AVERAGE PICK $e \simeq 2.718\ldots$ NUMBERS!

WE BEGIN BY DEFINING $N_x$ AS THE MIN. NUMBER OF VALUES ($U_k \in [0,1]$) THAT YOU NEED TO PICK SO THAT $\sum u_k > x$

$$N_x = \min\left\{n : \sum_{k=1}^{n} U_k > x\right\}$$

$$m_x = E(N_x)$$

WE WANT TO CALCULATE $m_1$

ASSUMING THAT $U_1 = u$, IF $u > x$ THEN $N_x = 1$ OTHERWISE IF $u < x$ $N_x = 1 + N'$ WHERE $N'$ IS DISTRIBUTED LIKE $N_{x-u}$. AS A RESULT

$$m_x = 1 + \int_0^x m_{x-u}\,du = 1 + \int_0^x m_u\,du \quad \underline{\text{IF } x \le 1}$$

NOW IT'S NOT DIFFICULT TO SEE THAT IF $\underline{m_x = e^x}$ THE EQUATION IS SATISFIED.

FINALLY WE HAVE $\boxed{m_1 = E(N_1) = e}$

@FERMATSLIBRARY

# Experiment with $e$ number


We can try theoretical blackboard solutions...

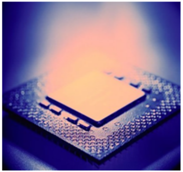
# Experiment with $e$ number


In many cases is very hard to obtain the theoretical solution...

# Experiment simulation with $e$ number

**But there are other strategies...**

# Experiment simulation with $e$ number



We can simulate this process on a computer!

# Experiment simulation with $e$ number

Much better approach in many cases!

# Experiment simulation with $e$ number


## Euler function

```
> sim_euler <- function(n){
+   m <- 0
+   for (i in 1:n) {
+     suma <- 0
+     while (suma < 1){
+       m <- m + 1
+       suma <- suma + runif(1, min = 0, max = 1)
+       }
+     }
+   return(m/n)
+ }
```

# Experiment simulation with $e$ number

Let's make the experiment 1000000 times.

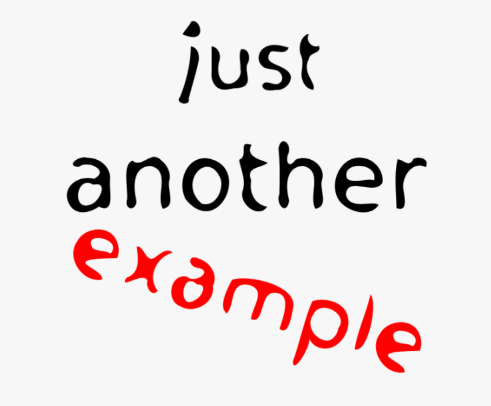
## Euler function

```
> time_1 <- proc.time()
> sim_euler(1000000)
[1] 2.71688
> proc.time() - time_1
      usuari      sistema transcorregut
        2.02         0.00          2.08
```

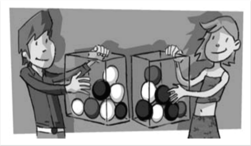
We compare with the theoretical value ...

## Exponential function

```
> exp(1)
[1] 2.718282
```


just
another
example

# Taking balls from an urn



You choose a ball at random in a urn with three balls labeled $\{0, 1, 2\}$. If the ball is 0 you are free. Otherwise, if the ball is $i \in \{1, 2\}$, you have to pay $i$ €, replace the ball and withdraw one again. Let X be the amount X of euros that you pay before being free. Compute the $E(X)$.

I had problems with this question of the exam! :)

# Taking balls from an urn

We can compute the analytical solution...

*Solution:*

Let $Y$ be the label of the ball in the first withdrawal. For each $k \geq 0$ we have

$$\mathrm{P}(X=k) = \mathrm{P}(X=k|Y=0)\mathrm{P}(Y=0) + \mathrm{P}(X=k|Y=1)\mathrm{P}(Y=1) + \mathrm{P}(X=k|Y=2)\mathrm{P}(Y=2).$$

We have

$$\mathrm{P}(X=k|Y=0) = \begin{cases} 1 & k=0 \\ 0 & k>0 \end{cases},$$

and

$$\mathrm{P}(X=k|Y=1) = \begin{cases} 0 & k=0 \\ \mathrm{P}(X=k-1) & k>0 \end{cases},$$

and

$$\mathrm{P}(X=k|Y=2) = \begin{cases} 0 & k\leq 1 \\ \mathrm{P}(X=k-2) & k>1 \end{cases}.$$

Hence,

$$\mathrm{P}(X=k) = \begin{cases} 1/3 & k=0 \\ 1/9 & k=1 \\ (1/3)(\mathrm{P}(X=k-1)+\mathrm{P}(X=k-2)) & k\geq 2 \end{cases}$$

# Taking balls from an urn

The generating function $G_X(t)$ satisfies

$$\begin{aligned}
G_X(t) &= \sum_{k\geq 0} \mathrm{P}(X=k)t^k \\
&= \frac{1}{3} + \frac{1}{9}t + \frac{1}{3}\sum_{k\geq 2}(\mathrm{P}(X=k-1)+\mathrm{P}(X=k-2))t^k \\
&= \frac{1}{3} + \frac{1}{9}t + \frac{1}{3}(\sum_{k\geq 2}\mathrm{P}(X=k-1)t^k) + \sum_{k\geq 2}\mathrm{P}(X=k-2)t^k) \\
&= \frac{1}{3} + \frac{1}{9}t + \frac{1}{3}(t(G_X(t)-1/3)+t^2G_X(t)) \\
&= \frac{1}{3} + \frac{1}{3}G_X(t)(t+t^2),
\end{aligned}$$

which implies

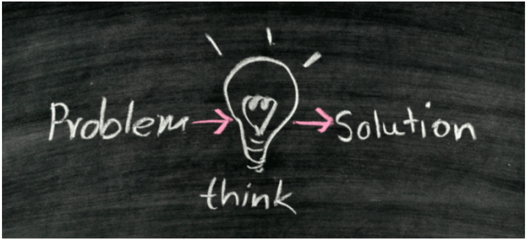
$$G_X(t) = \frac{1}{3-t-t^2}.$$

We have that

$$\mathrm{E}(X) = G'_X(1) = \left.\frac{2t+1}{(3-t-t^2)^2}\right|_{t=1} = 3.$$

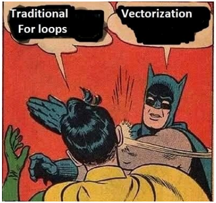
Very easy problem with “complex” theoretical solution!

# Taking balls from an urn




But we can compute this expectation empirically with resampling this process in a computer.

# Taking balls from an urn

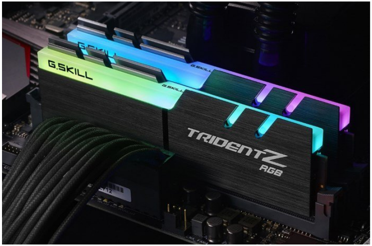



## Vectorized simulation

```
> time_2 <- proc.time()
> urn_simulation <- sample(c(0,1,2), 1000000, replace=TRUE, prob = c(1/3,1/3,1/3))
> sum(urn_simulation)/sum(urn_simulation==0)
[1] 3.004183
> proc.time() - time_2
     usuari     sistema transcorregut
       0.02        0.00          0.03
```

# Warning in vectorized computations

With vectorized computations maybe we are using much memory!

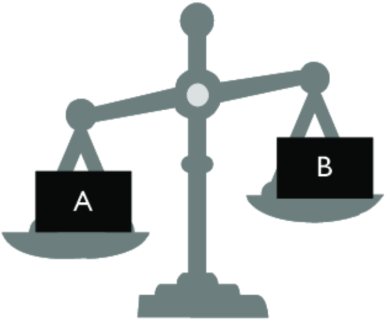




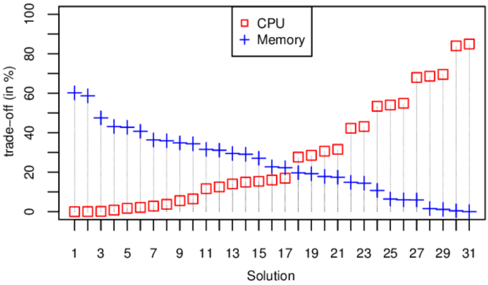
In the example above we have used 1,000,000 memory locations!
If you vectorize in a long code, remember to delete the variables you don't need...

# Computer Science trade-off

A **space-time trade off or time-memory trade-off**...

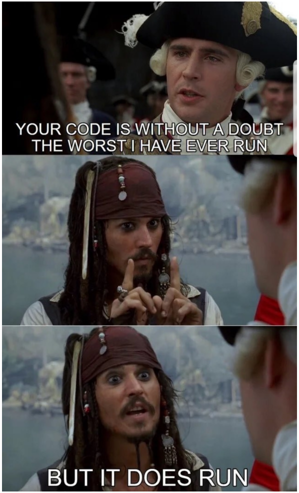


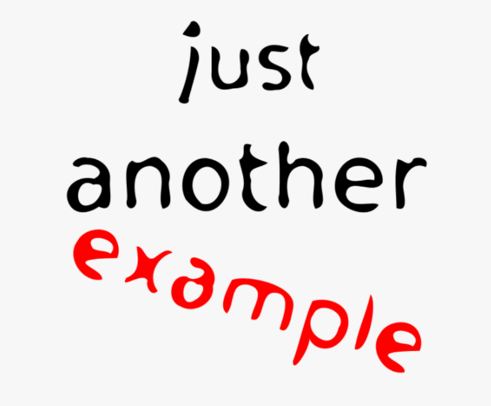
# Computer Science trade-off




... in computer science is a case where an algorithm or program trades increased space usage with decreased time. Here, space refers to the data storage consumed in performing a given task (RAM, HDD, etc), and time refers to the time consumed in performing a given task.

WIKIPEDIA


YOUR CODE IS WITHOUT A DOUBT
THE WORST I HAVE EVER RUN
BUT IT DOES RUN


just
another
example

# The anniversary problem


**The anniversary problem is a famous problem of probability and statistics**. The purpose of this problem is to determine the probability that there is a group of $n$ people that at least two coincide $(c = 2)$ on the date of birth (this is understood day and month), taking into account (obviously!) which is 365 days in the year.

# The anniversary problem

- Calculate the distribution function of this random variable for $c = 2$ with $n = 1 \ldots 100$.

# The anniversary problem

This process can be generalized to a group of $n$ people, where $p(n)$ is the probability of at least two of the $n$ people sharing a birthday. It is easier to first calculate the probability $\overline{p}(n)$ that all $n$ birthdays are *different*. According to the pigeonhole principle, $\overline{p}(n)$ is zero when $n > 365$. When $n \le 365$:

$$\begin{aligned}\bar{p}(n) &= 1 \times \left(1 - \frac{1}{365}\right) \times \left(1 - \frac{2}{365}\right) \times \cdots \times \left(1 - \frac{n-1}{365}\right) \\ &= \frac{365 \times 364 \times \cdots \times (365 - n + 1)}{365^n} \\ &= \frac{365!}{365^n (365-n)!} = \frac{n! \cdot \binom{365}{n}}{365^n}\end{aligned}$$

where ! is the factorial operator, $\binom{365}{n}$ is the binomial coefficient

The equation expresses the fact that the first person has no one to share a birthday, the second person cannot have the same birthday as the first ($\frac{364}{365}$), the third cannot have the same birthday as either of the first two ($\frac{363}{365}$), and in general the $n$th birthday cannot be the same as any of the $n-1$ preceding birthdays.

The event of at least two of the $n$ persons having the same birthday is complementary to all $n$ birthdays being different. Therefore, its probability $p(n)$ is

$$p(n) = 1 - \bar{p}(n).$$

# The anniversary problem

We can compute the analytical solution...

## Anniversary problem probabilities

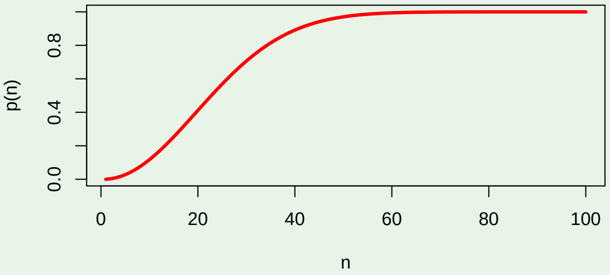
```
> n <- 1:100
> p_n <- 1 - (factorial(n) * choose(365, n)) / (365^n)
> p_n
  [1] 0.000000000 0.002739726 0.008204166 0.016355912 0.027135574 0.040462484
  [7] 0.056235703 0.074335292 0.094623834 0.116948178 0.141141378 0.167024789
 [13] 0.194410275 0.223102512 0.252901320 0.283604005 0.315007665 0.346911418
 [19] 0.379118526 0.411438384 0.443688335 0.475695308 0.507297234 0.538344258
 [25] 0.568699704 0.598240820 0.626859282 0.654461472 0.680968537 0.706316243
 [31] 0.730454634 0.753347528 0.774971854 0.795316865 0.814383239 0.832182106
 [37] 0.848734008 0.864067821 0.878219664 0.891231810 0.903151611 0.914030472
 [43] 0.923922856 0.932885369 0.940975899 0.948252843 0.954774403 0.960597973
 [49] 0.965779609 0.970373580 0.974431993 0.978004509 0.981138113 0.983876963
 [55] 0.986262289 0.988332355 0.990122459 0.991664979 0.992989448 0.994122661
 [61] 0.995088799 0.995909575 0.996604387 0.997190479 0.997683107 0.998095705
 [67] 0.998440043 0.998726391 0.998963666 0.999159576 0.999320753 0.999452881
 [73] 0.999560806 0.999648644 0.999719878 0.999777437 0.999823779 0.999860955
 [79] 0.999890668 0.999914332 0.999933109 0.999947953 0.999959646 0.999968822
 [85] 0.999975997 0.999981587 0.999985925 0.999989280 0.999991865 0.999993848
 [91] 0.999995365 0.999996521 0.999997398 0.999998061 0.999998560 0.999998935
 [97] 0.999999215 0.999999424 0.999999578 0.999999693
```

# The anniversary problem

And here we have the analitical plot solution...

## Anniversary problem probabilities

```
> plot(p_n, type='l', lwd=3, col='red', xlab='n', ylab='p(n)')
```



# Stochastic process definition

# Stochastic process definition

Stochastic (or random) process: a mathematical model of a magnitude that evolves as time passes in such a way that for any fixed instant of time one has a random variable.

**Definition**

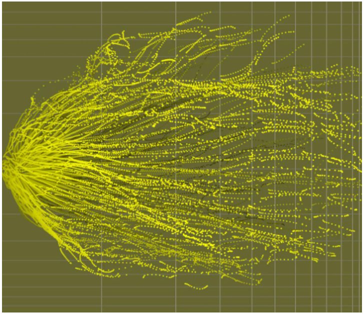
Let $(\Omega, \mathcal{F}, \mathrm{P})$ be a probability space and let $R_t \subset \mathbb{R}$.

A stochastic process $X$ is a mapping

$$\begin{aligned} X: \quad & R_t \times \Omega \longrightarrow \mathbb{R} \\ & (t, \omega) \mapsto X(t, \omega) \end{aligned}$$

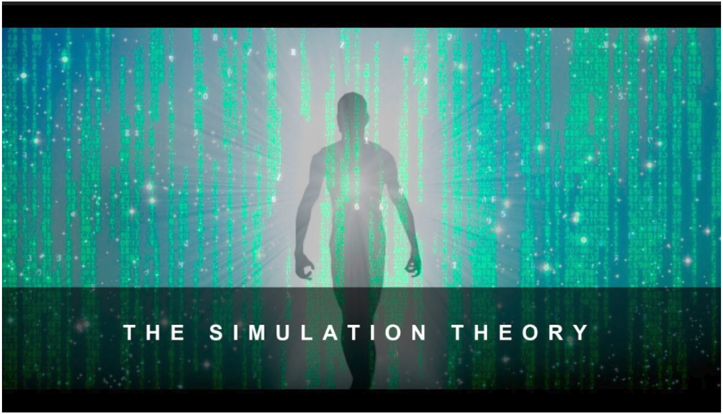
such that $X(t, \omega)$ is a random variable for all $t \in R_t$.

# Stochastic process



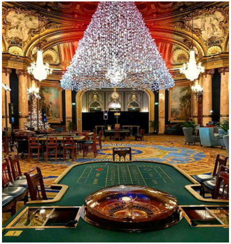
Resampling to make calculations (and understand!) **stochastic processes** like random walks or many others! They can be very complex and with analytical methods we cannot solve certain problems or computational methods allow us to obtain solutions more easily.

# Monte Carlo method



# Monte Carlo method

Probability simulation method...



- **Monte Carlo** methods (or **Monte Carlo experiments**) are a broad class of computational algorithms that rely on repeated random sampling to obtain numerical results.
- Their essential idea is using randomness to solve problems (stochastic or deterministic!).

# Monte Carlo method

- **In principle, Monte Carlo methods can be used to solve any problem having a probabilistic interpretation**.
- **They are often used in mathematical problems and are most useful when it is difficult or impossible to use other approaches**.
- Monte Carlo methods are used in many problem classes like **optimization, numerical integration, and generating draws from a probability distribution and many others**.
- By the **law of large numbers**, the expected value of some random variable can be approximated by taking the empirical mean of independent samples of the variable.

# The law of large numbers

In probability theory, the **law of large numbers (LLN)** is a theorem that describes the result of performing the same experiment a large number of times. According to the law, the average of the results obtained from a large number of trials should be close to the expected value, and will tend to become closer as more trials are performed.

**Theorem (Strong law)**

The **strong law of large numbers** states that the sample average converges almost surely to the expected value.

$$\overline{X}_n \to \mu \quad a.s. \ when \ n \to \infty$$

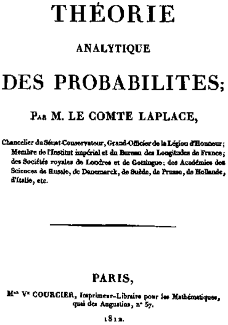
or

$$P\left(\lim \overline{X}_n = \mu\right) = 1 \quad when \ n \to \infty$$

# Monte Carlo method

Monte Carlo method was invented in the 1940s by Stanislaw Ulam, while he was working on nuclear weapons projects at the Los Alamos National Laboratory.

# Monte Carlo method


THÉORIE

ANALYTIQUE

DES PROBABILITES;

PAR M. LE COMTE LAPLACE,

Chancelier du Sénat-Conservateur, Grand-Officier de la Légion d'Honneur; Membre de l'Institut impérial et du Bureau des Longitudes de France; des Sociétés royales de Londres et de Gottingue; des Académies des Sciences de Russie, de Danemarck, de Suède, de Prusse, de Hollande, d'Italie, etc.

PARIS,

Mme Ve COURCIER, Imprimeur-Libraire pour les Mathématiques, quai des Augustins, n° 57.

1812.

$$P(A) = \frac{n}{N} = \frac{\#\,outcomes\ in\ A}{\#\,outcomes\ in\ Sample\ Space}$$

When Ulam was ill, he wanted to calculate the probability of finishing a game of solitaire. But he realized that it was something difficult to calculate. Then he thought of simulating the game on a computer.

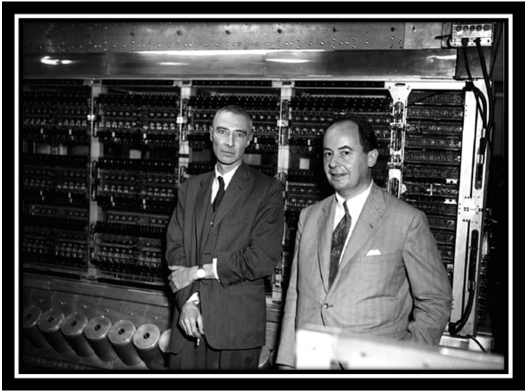
# Monte Carlo method

Stanislaw Ulam (1909-1984)

*"It began to take concrete form and began to develop with all its faults of rudimentary theory after I proposed it to Johnny!"*

# Monte Carlo method

Immediately after Ulam's breakthrough, John von Neumann understood its importance and programmed the ENIAC computer to carry out Monte Carlo calculations.

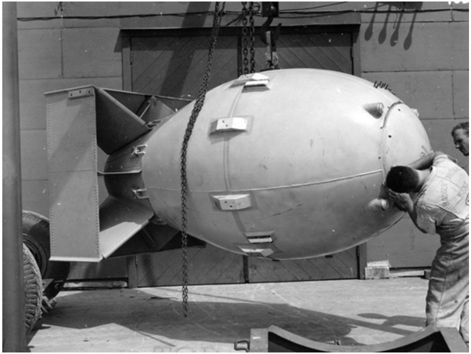


# Monte Carlo method

Montecarlo Casino

# Monte Carlo method

In 1940's, physicists at Los Alamos Scientific Laboratory were investigating radiation shielding and the distance that neutrons would likely travel through various materials.



# Monte Carlo method

Von Neumann

he contributed to the development of the Monte Carlo method, which allowed solutions to complicated problems to be approximated using random numbers.

Ulam

Metropolis

Paper 1953:
They applied MCMC for a chemical problem

Paper 1949: Using Markov chain for Monte Carlo approximation

Rossenburg

Teller

# Monte Carlo method


Monte Carlo methods vary, but tend to follow a particular pattern:

1. **Define a domain of possible inputs**.
2. **Generate many inputs randomly** from a probability distribution over the domain (resampling the process).
3. Perform a **deterministic computation** on the inputs.
4. **Aggregate the results**.

# The anniversary problem simulation

# The anniversary problem simulation

We can compute this with resampling simulation... with 100 simulations:

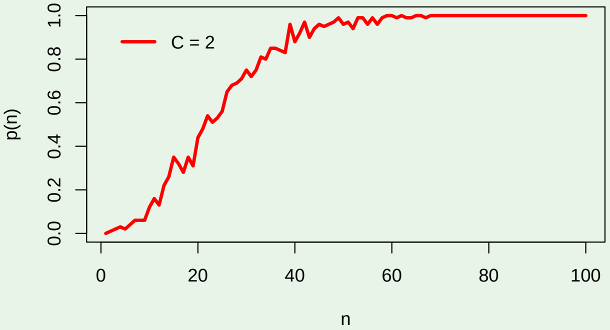
## Anniversary problem probabilities function

```
> n <- 100
> nsim <- 100
> p2_j <- 0
> for (i in 1:n) {
+   p2 <- 0
+   for (j in 1:nsim) {
+     taula_sim <- table(sample(1:365, i, repl=T))
+       if(any(taula_sim>1)) {
+           p2 <- p2+1
+         }
+     }
+   p2_j[i] <- p2/nsim
+ }
```

# The anniversary problem simulation

## Estimated probabilities plot with 100 simulations

```
> plot(p2_j, type='l', lwd=3, col='red', xlab='n', ylab='p(n)')
> legend(1, 0.98, legend=c("C = 2"), col=c("red"), lwd=3, box.lty=0)
```

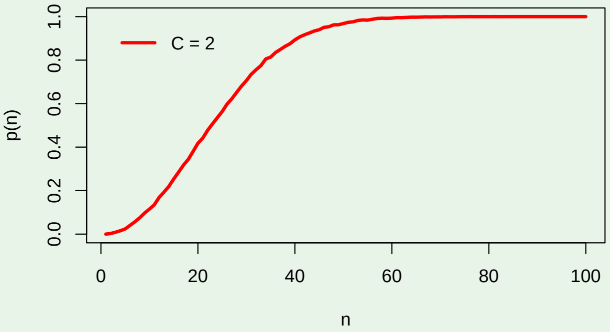


# The anniversary problem simulation

And with 10000 simulations:

## Estimated probabilities plot with 10000 simulations

```
> plot(p2_j, type='l', lwd=3, col='red', xlab='n', ylab='p(n)')
> legend(1, 0.98, legend=c("C = 2"), col=c("red"), lwd=3, box.lty=0)
```



# The anniversary problem simulation

And we can obtain the probabilities...

## Estimated probabilities

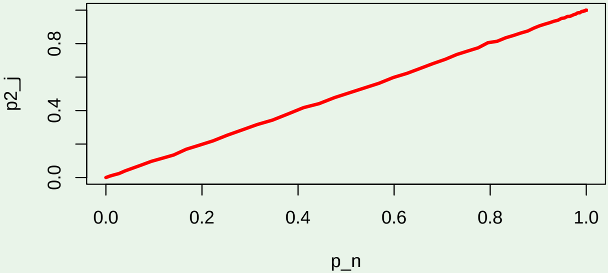
```
> p2_j
  [1] 0.0000 0.0028 0.0086 0.0155 0.0237 0.0402 0.0565 0.0752 0.0968 0.1149
 [11] 0.1348 0.1687 0.1931 0.2192 0.2532 0.2844 0.3167 0.3432 0.3799 0.4177
 [21] 0.4414 0.4773 0.5067 0.5353 0.5633 0.5976 0.6224 0.6521 0.6808 0.7060
 [31] 0.7350 0.7558 0.7750 0.8055 0.8139 0.8348 0.8491 0.8636 0.8754 0.8932
 [41] 0.9068 0.9167 0.9251 0.9339 0.9397 0.9506 0.9535 0.9622 0.9627 0.9682
 [51] 0.9738 0.9762 0.9824 0.9848 0.9839 0.9879 0.9916 0.9927 0.9921 0.9930
 [61] 0.9955 0.9953 0.9963 0.9976 0.9974 0.9980 0.9989 0.9985 0.9988 0.9988
 [71] 0.9995 0.9993 0.9992 0.9997 0.9997 1.0000 0.9997 0.9999 0.9999 0.9998
 [81] 1.0000 0.9998 0.9999 1.0000 0.9999 1.0000 1.0000 1.0000 1.0000 1.0000
 [91] 1.0000 1.0000 1.0000 1.0000 1.0000 1.0000 1.0000 1.0000 1.0000 1.0000
```

# The anniversary problem simulation

You can check your results comparing the theoretical distribution and the empirical one!

## Correlation between Theoretical and Empirical results

```
> plot(p_n, p2_j, type='l', lwd=3, col='red')
```



# The anniversary problem simulation



With resampling, we can easily compute a more complicated case!

# The anniversary problem simulation

- What are the probabilities for n = 30 and n = 100 with values? of c = 2, 3, 4? What is the *n* so that the probability is greater than 50%? And so that it is greater than 90%?

# The anniversary problem simulation

## Anniversary problem probabilities function with C = 2,3,4

```
> time_3 <- proc.time()
> n <- 100
> nsim <- 10000
> p2_j <- 0
> p3_j <- 0
> p4_j <- 0
> for (i in 1:n) {
+   p2 <- 0
+   p3 <- 0
+   p4 <- 0
+   for (j in 1:nsim) {
+     taula_sim <- table(sample(1:365, i, repl=T))
+       if(any(taula_sim>1)) {p2 <- p2+1}
+       if(any(taula_sim>2)) {p3 <- p3+1}
+       if(any(taula_sim>3)) {p4 <- p4+1}
+   }
+   p2_j[i] <- p2/nsim
+   p3_j[i] <- p3/nsim
+   p4_j[i] <- p4/nsim
+ }
> proc.time() - time_3

       usuari       sistema transcorregut
       123.84          2.36        128.60
```

# The anniversary problem simulation

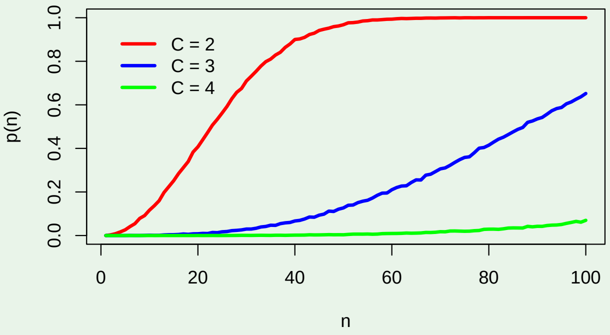
## Estimated probabilities

```
> p2_j[1:50]
 [1] 0.0000 0.0031 0.0083 0.0167 0.0261 0.0411 0.0544 0.0786 0.0915 0.1163
[11] 0.1367 0.1597 0.1974 0.2244 0.2516 0.2842 0.3118 0.3396 0.3832 0.4072
[21] 0.4398 0.4729 0.5071 0.5337 0.5627 0.5928 0.6281 0.6574 0.6752 0.7095
[31] 0.7313 0.7539 0.7781 0.7982 0.8101 0.8286 0.8413 0.8637 0.8796 0.9001
[41] 0.9026 0.9098 0.9229 0.9290 0.9415 0.9473 0.9521 0.9590 0.9621 0.9678
> p3_j[1:50]
 [1] 0.0000 0.0000 0.0000 0.0000 0.0000 0.0004 0.0002 0.0002 0.0006 0.0013
[11] 0.0008 0.0010 0.0027 0.0036 0.0038 0.0047 0.0070 0.0055 0.0077 0.0080
[21] 0.0097 0.0092 0.0140 0.0131 0.0170 0.0184 0.0223 0.0237 0.0258 0.0297
[31] 0.0298 0.0333 0.0393 0.0417 0.0473 0.0467 0.0546 0.0581 0.0601 0.0668
[41] 0.0695 0.0759 0.0850 0.0840 0.0934 0.0982 0.1121 0.1104 0.1210 0.1267
> p4_j[1:50]
 [1] 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000 0.0000
[11] 0.0000 0.0000 0.0001 0.0000 0.0000 0.0000 0.0001 0.0000 0.0000 0.0002
[21] 0.0000 0.0001 0.0002 0.0000 0.0001 0.0002 0.0003 0.0005 0.0008 0.0007
[31] 0.0005 0.0008 0.0011 0.0008 0.0006 0.0013 0.0013 0.0008 0.0014 0.0018
[41] 0.0019 0.0021 0.0033 0.0028 0.0029 0.0033 0.0040 0.0035 0.0037 0.0034
```

# The anniversary problem simulation

## Estimated probabilities plot

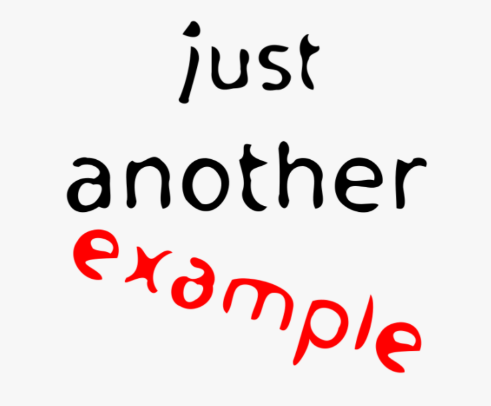
```
> plot(p2_j, type='l', lwd=3, col='red', xlab='n', ylab='p(n)')
> lines(p3_j, lwd=3, col='blue')
> lines(p4_j, lwd=3, col='green')
> legend(1,0.98,legend=c("C = 2","C = 3","C = 4"),col=c("red","blue","green"),lwd=3,box.lty=0)
```



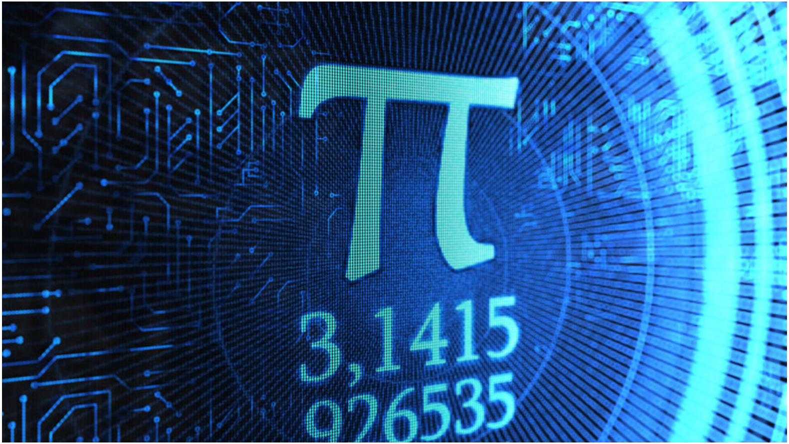
# The anniversary problem simulation

## Estimated probabilities in a particular case

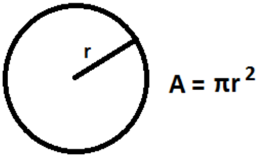
```
> which(p2_j>0.5)[1]; which(p2_j>0.9)[1]
[1] 23
[1] 40
> p2_j[30]; p2_j[100]
[1] 0.7095
[1] 1
> p3_j[30]; p3_j[100]
[1] 0.0297
[1] 0.6521
> p4_j[30]; p4_j[100]
[1] 7e-04
[1] 0.0698
```


just
another
example

# The $\pi$ aproximation



# The $\pi$ aproximation




In this procedure the domain of inputs is the square that circumscribes the quadrant. We generate random inputs by scattering grains over the square then perform a computation on each input (test whether it falls within the quadrant). Aggregating the results yields our final result, the approximation of $\pi$.

## The $\pi$ aproximation

For example, consider a quadrant (circular sector) inscribed in a unit square. Given that the ratio of their areas is $\pi/4$, the value of $\pi$ can be approximated using a Monte Carlo method:

1. Draw a square, then inscribe a quadrant within it.
2. Uniformly scatter a given number of points over the square.
3. Count the number of points inside the quadrant, i.e. having a distance from the origin of less than 1.
4. The ratio of the inside-count and the total-sample-count is an estimate of the ratio of the two areas, $\pi/4$. Multiply the result by 4 to estimate $\pi$.

# The $\pi$ aproximation

In this procedure the domain of inputs is the square that circumscribes the quadrant. We generate random inputs by scattering grains over the square then perform a computation on each input (test whether it falls within the quadrant). Aggregating the results yields our final result, the approximation of $\pi$.

# The $\pi$ aproximation

## function

```
> runs <- 10000
> #runif samples from a uniform distribution
> xs <- runif(runs, min = 0, max= 1)
> ys <- runif(runs, min = 0, max = 1)
> in_circle <- (xs^2 + ys^2) <= 1
> mc_pi <- (sum(in_circle)/runs)*4
> mc_pi
[1] 3.128
```

$\pi$ estimation...

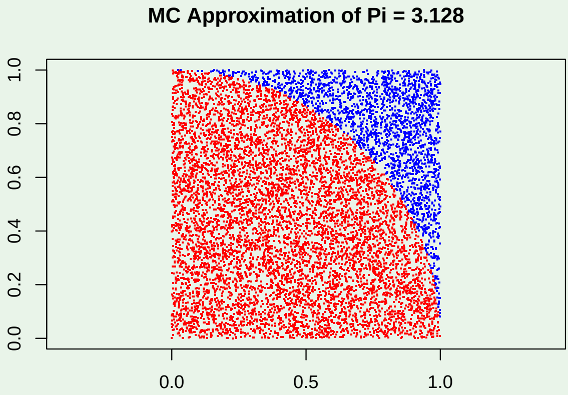
## function

```
> mc_pi
[1] 3.128
```

# The $\pi$ aproximation

## function

```
> plot(xs, ys, pch = '.', col = ifelse(in_circle, "red", "blue"), xlab='', ylab='',
+       asp = 1, main = paste("MC Approximation of Pi =", mc_pi))
```



# Monte Carlo method tips and tricks

- If the points are not uniformly distributed, then the approximation will be poor (or bad!).
- There are a large number of points. The approximation is generally poor if only a few points are randomly placed in the whole square. On average, the approximation improves as more points are placed.
- Uses of Monte Carlo methods require large amounts of random numbers, and it was their use that spurred the development of pseudorandom number generators, which were far quicker to use than the tables of random numbers that had been previously used for statistical sampling.

# Limitations of Monte Carlo methods

A limitation of the method is that **can be slow**. Refinements of the method exist to apply them in certain particular cases and accelerate the speed of convergence. **It is very general, but this leads to limitations in the speed of convergence.**

# References

- Christian P. Robert Good; George Casella *Introducing Monte Carlo Methods with R*, Springer, (2010).
- Christian P. Robert Good; George Casella *Monte Carlo Statistical Methods*, Springer, (2004).


The End